Extra Dessert #12: Frequency Privileges and Band Edges

Carrier-frequency vs sideband-edge math, 60 m channelized rules, 2200 m and 630 m power limits, ships and aircraft.

Group E1A · 11 pool questions · Self-study reference · Practice exams at eham.net/exams

Carrier-Frequency vs Sideband Math (HF Phone)

On HF SSB, the dial shows the **suppressed carrier** frequency. The actual transmitted signal occupies a band of frequencies above (USB) or below (LSB) the carrier. Sideband edges must stay inside the authorized band segment.

USB: signal extends from carrier up to (carrier + audio max). So the highest legal USB carrier = upper band edge − max audio frequency.
LSB: signal extends from (carrier − audio max) up to carrier. So the lowest legal LSB carrier = lower band edge + max audio frequency.

Worked examples

Question	Answer	Q
Why is 3 kHz USB at 14.348 MHz not legal?	**The upper 1 kHz of the signal is outside the 20-meter band** (14.348 + 3 = 14.351, but band ends at 14.350)	E1A01
Lowest carrier frequency for a 3 kHz LSB signal totally inside the band	**3 kHz above the lower band edge**	E1A02
Highest carrier for 2.8 kHz USB data on 20 m	**14.1472 MHz** (data segment ends at 14.150; 14.150 − 0.0028 = 14.1472)	E1A03
May an Extra answer LSB CQ on 3.601 MHz?	**No, the sideband components will extend beyond the edge of the phone band segment** (phone segment begins at 3.600; LSB extends below the carrier into 3.598 territory)	E1A04

60 m Channelized Operation

Question	Answer	Q
Required CW transmit frequency on 60 m	**At the center frequency of the channel**	E1A06

2200 m and 630 m Power Limits

Band	Max power	Q
2200 m	**1 watt EIRP**	E1A07
630 m (except parts of Alaska)	**5 watts EIRP**	E1A09

Ships, Aircraft, And Message Forwarding

Question	Answer	Q
Who must control an amateur station on a US-documented vessel/aircraft?	**Any person holding an FCC-issued amateur license or who is authorized for alien reciprocal operation**	E1A05
Licensing for ham operation on a US vessel in international waters	**Any FCC-issued amateur license**	E1A11
Condition before operating an amateur station on a ship or aircraft	**Operation must be approved by the master of the ship or the pilot in command of the aircraft**	E1A10
Who is primarily accountable if a forwarding station relays a violating message?	**The control operator of the originating station**	E1A08

Dessert 12 Question Bank (E1A)

E1A01. Why is it not legal to transmit a 3 kHz bandwidth USB signal with a carrier frequency of 14.348 MHz?
A. USB is not used on 20-meter phone.
B. The lower 1 kHz of the signal is outside the 20-meter band.
C. 14.348 MHz is outside the 20-meter band.
D. The upper 1 kHz of the signal is outside the 20-meter band.

E1A02. When using a transceiver that displays the carrier frequency of phone signals, which of the following displayed frequencies represents the lowest frequency at which a properly adjusted LSB emission will be totally within the band?
A. The exact lower band edge.
B. 300 Hz above the lower band edge.
C. 1 kHz above the lower band edge.
D. 3 kHz above the lower band edge.

E1A03. What is the highest legal carrier frequency on the 20-meter band for transmitting a 2.8 kHz wide USB data signal?
A. 14.0708 MHz.
B. 14.1002 MHz.
C. 14.1472 MHz.
D. 14.3490 MHz.

E1A04. May an Extra class operator answer the CQ of a station on 3.601 MHz LSB phone?
A. Yes, the entire signal will be inside the SSB allocation for Extra class operators.
B. Yes, the displayed frequency is within the 75-meter phone band segment.
C. No, the sideband components will extend beyond the edge of the phone band segment.
D. No, US stations are not permitted to use phone emissions below 3.610 MHz.

E1A05. Who must be in physical control of the station apparatus of an amateur station aboard any vessel or craft that is documented or registered in the United States?
A. Only a person with an FCC Marine Radio license grant.
B. Only a person named in an amateur station license grant.
C. Any person holding an FCC issued amateur license or who is authorized for alien reciprocal operation.
D. Any person named in an amateur station license grant or a person holding an unrestricted Radiotelephone Operator Permit.

E1A06. What is the required transmit frequency of a CW signal for channelized 60 meter operation?
A. At the lowest frequency of the channel.
B. At the center frequency of the channel.
C. At the highest frequency of the channel.
D. On any frequency where the signal's sidebands are within the channel.

E1A07. What is the maximum power permitted on the 2200-meter band?
A. 50 watts PEP (peak envelope power).
B. 100 watts PEP (peak envelope power).
C. 1 watt EIRP (equivalent isotropic radiated power).
D. 5 watts EIRP (equivalent isotropic radiated power).

E1A08. If a station in a message forwarding system inadvertently forwards a message that is in violation of FCC rules, who is primarily accountable for the rules violation?
A. The control operator of the packet bulletin board station.
B. The control operator of the originating station.
C. The control operators of all the stations in the system.
D. The control operators of all the stations in the system not authenticating the source from which they accept communications.

E1A09. Except in some parts of Alaska, what is the maximum power permitted on the 630-meter band?
A. 50 watts PEP (peak envelope power).
B. 100 watts PEP (peak envelope power).
C. 1 watt EIRP (equivalent isotropic radiated power).
D. 5 watts EIRP (equivalent isotropic radiated power).

E1A10. If an amateur station is installed aboard a ship or aircraft, what condition must be met before the station is operated?
A. Its operation must be approved by the master of the ship or the pilot in command of the aircraft.
B. The amateur station operator must agree not to transmit when the main radio of the ship or aircraft is in use.
C. The amateur station must have a power supply that is completely independent of the main ship or aircraft power supply.
D. The amateur station must operate only in specific segments of the amateur service HF and VHF bands.

E1A11. What licensing is required when operating an amateur station aboard a US-registered vessel in international waters?
A. Any amateur license with an FCC Marine or Aircraft endorsement.
B. Any FCC-issued amateur license.
C. Only General class or higher amateur licenses.
D. An unrestricted Radiotelephone Operator Permit.

Answer Key

E1A01: D	E1A02: D	E1A03: C	E1A04: C
E1A05: C	E1A06: B	E1A07: C	E1A08: B
E1A09: D	E1A10: A	E1A11: B	